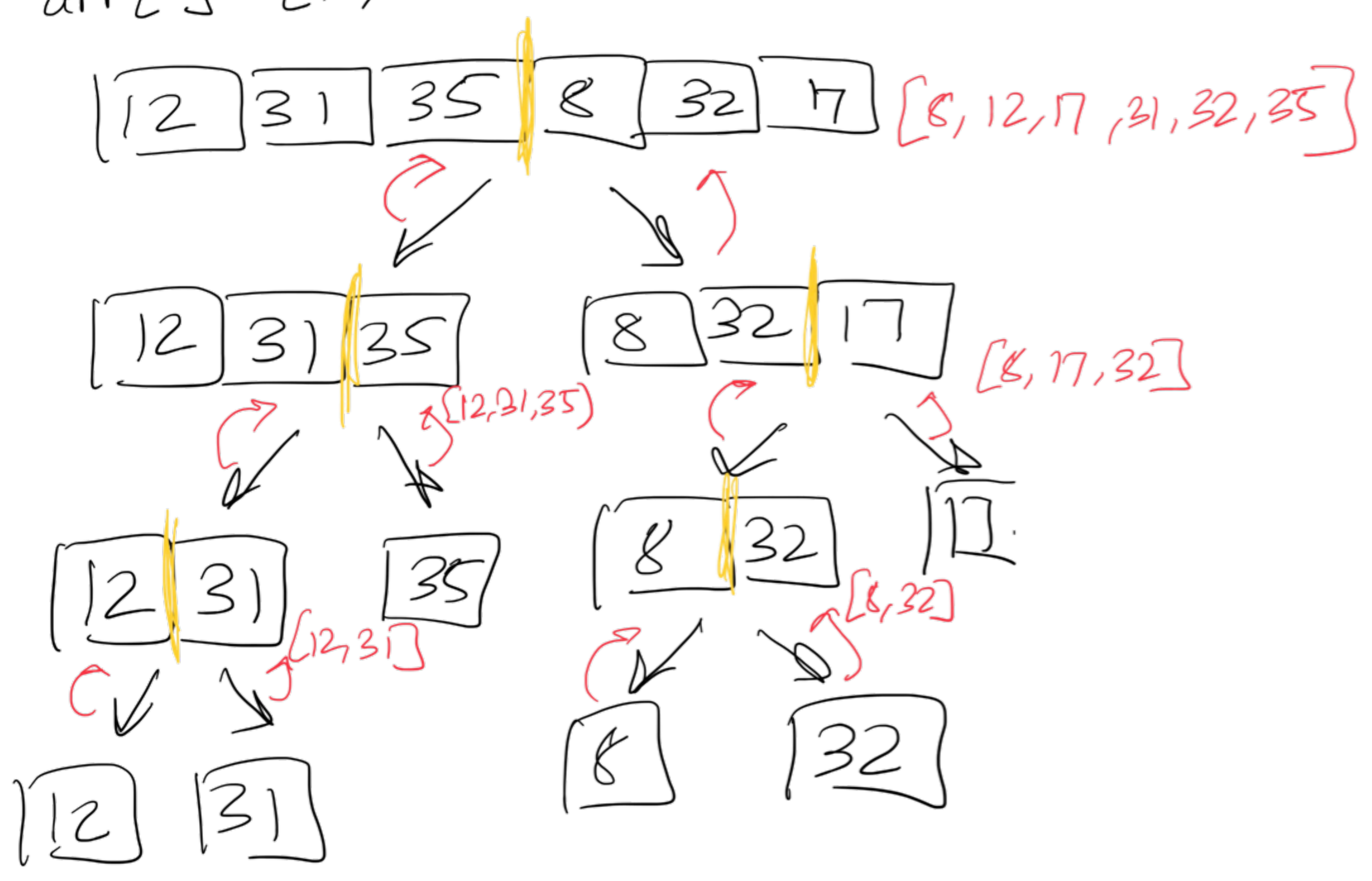
Merge Sort

arr[] = [12, 31, 35, 8, 32, 17]



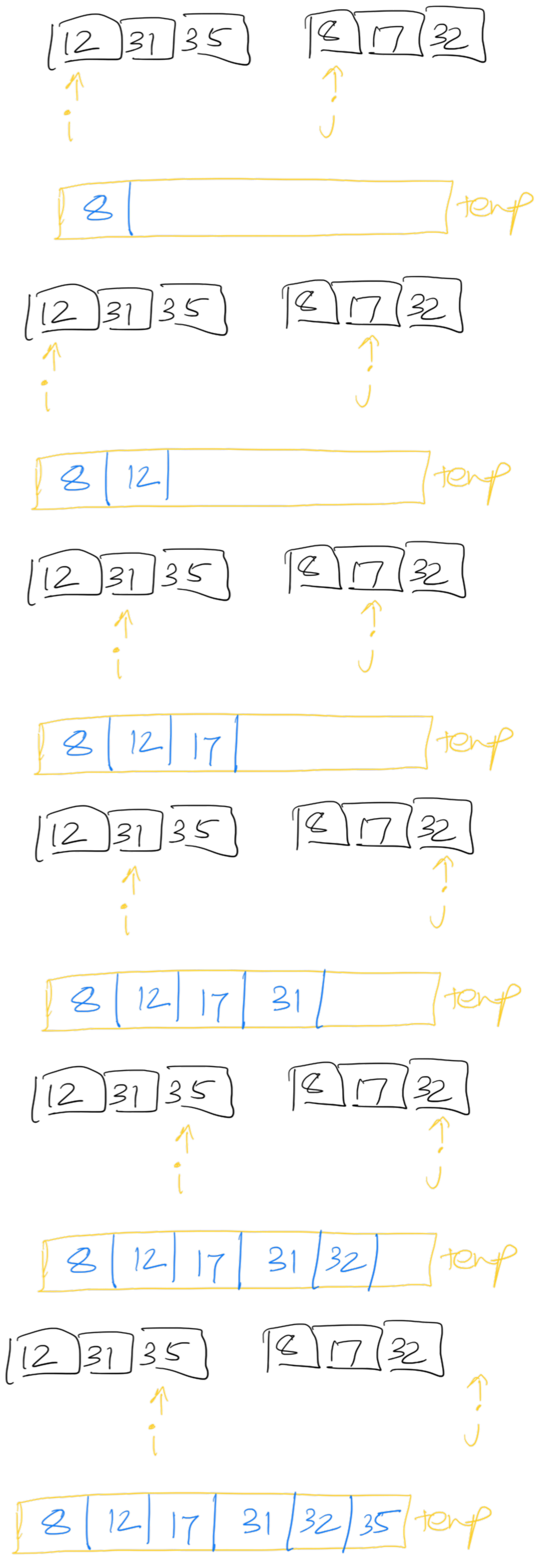
1. Divide
2. merge parts to create sorted array

Function to recursively divide

```
void mergeSort (arr[], start, end){
    while (s<e) {
        int mid = st + (end-st)/2
        mergeSort(arr, st, mid) //left half pura divide karega
        mergeSort (arr, mid+1, end) //fir right half divide kete jayege
    }
}
```

Function to merge

```
void merge (arr, st, mid, end) {
    vector<int> temp;
    i=st, j=mid+1;
    while(i<=mid && j<=end){
        if (A[i] <= A[j]) {
            temp.pb(A[i])
            i++;
        } else {
            temp.pb(A[j])
            j++
        }
    }
    //if elements still left in other
      array then run this loop to copy
    while (i<=mid) { //left half
        temp.pb(a[i])
        i++
    }
    while (j<=end) { //right
        temp.pb(a[j])
        j++
    }
    for (idx=0; idx<temp.size(); idx++){
        A[idx + st] = temp[idx]
    }
}
```



if [5] [3]

st=2 end=3

[3 | 5] temp

idx 0 1

so to copy back and find idx to paste in og array we will do idx + st and paste

then i++

AND COPY ELEMENTS OF TEMP IN OG ARRAY